

Structural Gravity, from Theory to Practice

A fully derived study companion to Chapter 1 of the UNCTAD–WTO
Advanced Guide to Trade Policy Analysis: The Structural Gravity Model
(Yotov, Piermartini, Monteiro and Larch, 2016)
with lesson quizzes and a final exam

Study notes

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Part I

The Structural Gravity Model: From Theory to a Testable Equation

Trade economists call the gravity equation the “workhorse” of the field: bilateral trade between two countries rises with the two countries’ economic size and falls with the frictions between them, in close analogy with Newton’s law of universal gravitation. Unlike the toy Ricardian and DFS models built from scratch in earlier study companions, the gravity model was for decades an atheoretical regularity (Tinbergen, 1962) before Anderson (1979), Anderson and van Wincoop (2003), and Eaton and Kortum (2002) gave it rigorous general-equilibrium foundations. This Part rebuilds that foundation — the *structural gravity model* — from a representative Armington–CES economy, term by term.

1 Lesson 1: The gravity analogy and the evolution of gravity theory

1.1 Newton’s law, applied to trade

Intuition: Particles that trade

Newton’s Law of Universal Gravitation states that the attractive force between two particles is proportional to the product of their masses and inversely proportional to the square of the distance between them. Ravenstein (1885) and Tinbergen (1962) noticed that bilateral trade (and migration) flows obey a strikingly similar pattern: big, close economies trade a lot; small, distant economies trade little. For forty years this was simply a good empirical regularity with no theoretical pedigree — an equation in search of a model.

Definition: Box 1 — The Newton/gravity analogy, formalized

Once the structural gravity model is derived (Lesson 2), the analogy becomes exact:

$$\underbrace{F_{ij} = G \frac{M_i M_j}{D_{ij}^2}}_{\text{Newton's Law}} \quad \longleftrightarrow \quad \underbrace{X_{ij} = G \frac{Y_i E_j}{T_{ij}^\theta}}_{\text{Gravity Trade Model}}$$

where X_{ij} is exports from i to j ; $G \equiv 1/Y$ is the inverse of world output (the “gravitational constant”); Y_i is country i ’s production (mass); E_j is country j ’s expenditure (mass); and $T_{ij}^\theta \equiv (t_{ij}(\Pi_i P_j))^{\sigma-1}$ packages *all* trade frictions between i and j , direct and structural, into a single “distance” term. The mapping is not a metaphor bolted on after the fact: it falls directly out of the algebra of Lesson 2.

1.2 From an atheoretical curve-fit to a family of structural models

Intuition: Many roads to the same gravity equation

Anderson (1979) first derived gravity from CES preferences with product differentiation by place of origin. Bergstrand (1985) offered an early alternative. But gravity only became central to trade theory once Eaton and Kortum (2002, “EK”) derived it on the *supply side* from a Ricardian model with a continuum of goods and Fréchet-distributed productivities, and Anderson and van Wincoop (2003, “AvW”) popularized the demand-side Armington-CES derivation and stressed the importance of general-equilibrium price terms. Arkolakis, Costinot and Rodríguez-Clare (2012) then showed something remarkable: a whole *family* of seemingly different trade models — Armington/monopolistic competition (Anderson, AvW), Heckscher–Ohlin (Bergstrand, Deardorff), Ricardian (EK), heterogeneous firms (Chaney; Helpman–Melitz–Rubinstein), sectoral and dynamic extensions — collapse to *isomorphic* gravity equations that deliver the same formula for the gains from trade. Gravity is therefore not one model but a common reduced form of an entire class of general-equilibrium trade models.

Understanding Quiz 1 — Lesson 1

Q1.1 In the Newton/gravity analogy, the term T_{ij}^θ corresponds to:

- (A) the product of the two countries’ GDPs
- (B) the square of physical distance only
- (C) all bilateral and structural trade frictions between i and j combined
- (D) the elasticity of substitution

Q1.2 Before Anderson (1979), the gravity equation in trade was mainly:

- (A) derived from the Ricardian model
- (B) an atheoretical empirical regularity
- (C) derived from Heckscher–Ohlin
- (D) rejected by the data

Q1.3 The main contribution of Arkolakis, Costinot and Rodríguez-Clare (2012) was to show that:

- (A) only the Armington model generates gravity
- (B) a wide class of structurally different trade models generate isomorphic gravity equations and identical formulas for the gains from trade
- (C) gravity does not hold with sectoral data
- (D) trade costs are irrelevant for welfare

2 Lesson 2: Deriving structural gravity from the demand side

2.1 Environment

Definition: The Armington–CES world

The world has N countries. Each country i produces a single, exogenously fixed quantity Q_i of its own differentiated variety (goods are differentiated *by place of origin*, following Armington, 1969) at factory-gate price p_i . Nominal output is $Y_i \equiv p_i Q_i$. Country j ’s

aggregate expenditure is E_j , related to its own income by $E_j = \phi_j Y_j$, where $\phi_j > 1$ denotes a trade deficit and $0 < \phi_j < 1$ a trade surplus (deficits/surpluses are treated as exogenous, as in Dekle, Eaton and Kortum, 2007–2008). Shipping a unit of i 's variety to j requires shipping $t_{ij} \geq 1$ units (an *iceberg* trade cost, Samuelson 1952): $1/t_{ij}$ of the shipment “melts” in transit, so the price consumers in j pay is $p_{ij} = p_i t_{ij}$.

Intuition: Why iceberg costs?

Modelling trade costs as “melting” cargo rather than as a separate transportation sector keeps the algebra one-good-per-origin instead of two. It is a modelling convenience, not a claim about how shipping actually works — and, as Anderson (2011) notes, the iceberg metaphor stretches naturally to *fixed* costs too (“a chunk of the iceberg breaks off as it leaves the mother glacier”).

Consumers in j have identical, homothetic CES preferences over varieties from every origin i :

Definition: CES preferences and budget constraint

$$U_j = \left(\sum_i \alpha_i^{1/\sigma} c_{ij}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1, \quad (1-1)$$

where σ is the elasticity of substitution across origins, $\alpha_i > 0$ is an exogenous CES taste parameter for origin i 's variety, and c_{ij} is j 's consumption of i 's variety. Consumers maximize (1-1) subject to

$$\sum_i p_{ij} c_{ij} = E_j, \quad p_{ij} = p_i t_{ij}. \quad (1-2)$$

2.2 Solving the consumer's problem

Derivation: Demand and the CES price index

Standard CES optimization of (1-1) subject to (1-2) yields bilateral expenditure

$$X_{ij} = \left(\frac{\alpha_i p_i t_{ij}}{P_j} \right)^{1-\sigma} E_j, \quad (1-3)$$

where the CES consumer price index is

$$P_j = \left(\sum_i (\alpha_i p_i t_{ij})^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \quad (1-4)$$

Intuition: Reading equation (1-3) term by term

Since $\sigma > 1$, expenditure on goods from i , X_{ij} , is:

- (i) **Proportional to E_j** : bigger/richer markets buy more of everything, including goods from i .

- (ii) **Inversely related to the delivered price** $p_{ij} = p_i t_{ij}$: the law of demand, sharpened by trade costs. The best combination for bilateral trade is a low factory-gate price *and* a low bilateral trade cost.
- (iii) **Directly related to P_j** : the more expensive the rest of the world's varieties are, the more j 's consumers substitute toward i .
- (iv) **Scaled by σ** : a higher elasticity of substitution *magnifies* trade diversion between cheap and expensive sources.

2.3 Closing the model: market clearing

Derivation: From bilateral demand to the structural gravity system

Market clearing requires that country i 's output, valued at delivered prices, equal total world spending on i 's variety:

$$Y_i = \sum_j \left(\frac{\alpha_i p_i t_{ij}}{P_j} \right)^{1-\sigma} E_j \equiv \sum_j X_{ij}. \quad (1-5)$$

Let $Y \equiv \sum_i Y_i$ be world output. Dividing (1-5) by Y and rearranging:

$$(\alpha_i p_i)^{1-\sigma} = \frac{Y_i/Y}{\sum_j (t_{ij}/P_j)^{1-\sigma} (E_j/Y)}. \quad (1-6)$$

Following Anderson and van Wincoop (2003), define the **outward multilateral resistance** Π_i implicitly by the denominator of (1-6):

$$\Pi_i^{1-\sigma} \equiv \sum_j \left(\frac{t_{ij}}{P_j} \right)^{1-\sigma} \frac{E_j}{Y} \implies (\alpha_i p_i)^{1-\sigma} = \frac{Y_i/Y}{\Pi_i^{1-\sigma}}. \quad (1-7)$$

Substituting (1-7) back into (1-3) and (1-4) delivers the celebrated **structural gravity system**:

$$X_{ij} = \frac{Y_i E_j}{Y} \left(\frac{t_{ij}}{\Pi_i P_j} \right)^{1-\sigma} \quad (1-8)$$

$$\Pi_i^{1-\sigma} = \sum_j \left(\frac{t_{ij}}{P_j} \right)^{1-\sigma} \frac{E_j}{Y} \quad P_j^{1-\sigma} = \sum_i \left(\frac{t_{ij}}{\Pi_i} \right)^{1-\sigma} \frac{Y_i}{Y}. \quad (1-9, 1-10)$$

Intuition: Why the system is genuinely *general* equilibrium

Equations (1-9) and (1-10) are not free-standing definitions: Π_i depends on every P_j in the world, and every P_j depends on every Π_i . A trade-cost shock *anywhere* ripples through the whole $\{\Pi_i, P_j\}$ system before it settles into new bilateral trade flows via (1-8). This is exactly the property that a naive, atheoretical gravity regression misses — and exactly what Chapter 2 of this Guide is built to quantify.

Understanding Quiz 2 — Lesson 2

Q2.1 In the Armington–CES structural gravity model, goods are differentiated:

- (A) by sector only
- (B) by place of origin
- (C) by firm
- (D) they are homogeneous

Q2.2 Bilateral expenditure X_{ij} in equation (1-3) is increasing in:

- (A) the delivered price p_{ij}
- (B) destination expenditure E_j and the destination price index P_j
- (C) bilateral trade costs t_{ij} only
- (D) the elasticity of substitution σ only

Q2.3 The outward multilateral resistance Π_i is defined so that:

- (A) it depends only on country i 's own trade costs
- (B) it aggregates, across all destinations j , the trade costs i faces weighted by destination expenditure shares
- (C) it is a exogenous parameter set by policy
- (D) it equals the elasticity of substitution

Q2.4 The multilateral resistance system (1-9)–(1-10) is “general equilibrium” because:

- (A) it can be solved country by country independently
- (B) Π_i and P_j are jointly, simultaneously determined for all countries at once
- (C) it ignores trade costs
- (D) it only holds in autarky

3 Lesson 3: Size vs. trade costs, and the supply-side (Eaton–Kortum) derivation

3.1 Decomposing gravity: size term and trade-cost term

Derivation: Splitting equation (1-8) in two

Equation (1-8) decomposes cleanly into a **size term** and a **trade-cost term**:

$$X_{ij} = \underbrace{\frac{Y_i E_j}{Y}}_{\text{size}} \times \underbrace{\left(\frac{t_{ij}}{\Pi_i P_j} \right)^{1-\sigma}}_{\text{trade cost}}.$$

Intuition: Reading the two terms

Size term $Y_i E_j / Y$: setting $t_{ij} = 1$ for all pairs recovers this term exactly — it is the hypothetical *frictionless* trade flow, i.e. what trade would be if geography and policy did not matter. In a frictionless world, j 's expenditure share on i equals i 's share of world output, $X_{ij} / E_j = Y_i / Y$. Large producers export more everywhere; large/rich markets import more from everywhere; and similarly sized partners trade relatively more with each other.

Trade-cost term $(t_{ij} / (\Pi_i P_j))^{1-\sigma}$: this is the wedge between realized and frictionless trade, built from (i) the observable bilateral cost t_{ij} (distance, tariffs, RTAs...); (ii) P_j , importer

j 's ease of market access (*inward* multilateral resistance); and (iii) Π_i , exporter i 's ease of market access (*outward* multilateral resistance). The multilateral resistances are precisely what translate a bilateral, partial-equilibrium trade-cost change into economy-wide, general-equilibrium consequences — the subject of Chapter 2.

3.2 Appendix A: the same gravity system from the supply side (Eaton–Kortum)

Intuition: Two roads, one destination

Section B.1 derived structural gravity from Armington–CES *demand*. Eaton and Kortum (2002) derive an isomorphic system from *supply*: countries differ not in tastes for differentiated varieties but in Ricardian productivity, drawn from a Fréchet distribution, for a continuum of homogeneous goods. The two derivations produce the identical reduced form, with $-\theta$ (the Fréchet dispersion parameter) playing the role of $1 - \sigma$.

Derivation: Sketch of the EK supply-side derivation

There is a continuum of goods $l \in [0, 1]$ and CES utility $U_j = (\int_0^1 c(l)^{\frac{\sigma-1}{\sigma}} dl)^{\sigma/(\sigma-1)}$. Country i 's efficiency at producing good l is $z_i(l)$, drawn from a Fréchet distribution $F_i(z) = e^{-T_i z^{-\theta}}$ (the same distribution derived from first principles as the limit of “best-of-many-Pareto-ideas” in the EK/Melitz companion). With constant returns and input cost ς_i , delivering good l from i to j costs $p_{ij}(l) = (\varsigma_i/z_i(l)) t_{ij}$, and competitive markets mean consumers pay the *minimum* price across all possible sources, $p_j(l) = \min_i \{p_{ij}(l)\}$. Standard extreme-value algebra (identical in spirit to the price-distribution derivation of Lesson 9 of the EK companion) gives the distribution of the price actually paid in j as

$$G_j(p) = 1 - e^{-\Phi_j p^\theta}, \quad \Phi_j \equiv \sum_i T_i (\varsigma_i t_{ij})^{-\theta},$$

and the probability (equivalently, by the Law of Large Numbers over the continuum, the *expenditure share*) that j buys good l from i is

$$\pi_{ij} = \frac{T_i (\varsigma_i t_{ij})^{-\theta}}{\Phi_j}, \quad X_{ij} = \pi_{ij} E_j.$$

Imposing market clearing $Y_i = \sum_j X_{ij}$ exactly as in Lesson 2, and defining Π_i and P_j analogously, delivers

$$X_{ij} = \frac{Y_i E_j}{Y} \left(\frac{t_{ij}}{\Pi_i P_j} \right)^{-\theta}, \quad \Pi_i^{-\theta} = \sum_j \left(\frac{t_{ij}}{P_j} \right)^{-\theta} \frac{E_j}{Y}, \quad P_j^{-\theta} = \sum_i \left(\frac{t_{ij}}{\Pi_i} \right)^{-\theta} \frac{Y_i}{Y},$$

term for term identical to (1-8)–(1-10) with $1 - \sigma$ replaced by $-\theta$.

Intuition: Why this matters

Because CES/Armington and Ricardian/EK deliver the *same* estimating equation, an econometrician cannot tell, from aggregate bilateral trade flows alone, which underlying model generated the data. This is a feature, not a bug: it means the gravity toolkit built in this Guide applies regardless of which structural story a researcher prefers, and it is the technical content behind the Arkolakis-Costinot-Rodríguez-Clare isomorphism result of Lesson 1.

Understanding Quiz 3 — Lesson 3

- Q3.1** The frictionless (size) component of the gravity equation, $Y_i E_j / Y$, is obtained by:
- (A) setting $\sigma = 1$
 - (B) setting all bilateral trade costs $t_{ij} = 1$
 - (C) ignoring expenditure
 - (D) assuming autarky
- Q3.2** The multilateral resistances Π_i and P_j enter the trade-cost term because they capture:
- (A) the size of country i and j only
 - (B) each country's ease of market access relative to the rest of the world
 - (C) random measurement error
 - (D) tariff revenue
- Q3.3** In the Eaton–Kortum supply-side derivation, the role played by $1 - \sigma$ in the CES model is played by:
- (A) $-\theta$, the Fréchet dispersion parameter
 - (B) the number of countries N
 - (C) the labour endowment
 - (D) the CES taste parameter α_i
- Q3.4** The practical implication of the demand-side/supply-side isomorphism is that:
- (A) gravity only holds under Armington preferences
 - (B) aggregate bilateral trade data alone cannot distinguish which structural model generated the gravity equation
 - (C) Ricardian trade models do not generate gravity
 - (D) trade costs are unidentified in both models

Part II

Estimating Structural Gravity: Eight Challenges and Six Recommendations

Log-linearizing (1-8) with an additive error term gives the workhorse empirical gravity equation

$$\ln X_{ij,t} = \ln E_{j,t} + \ln Y_{i,t} - \ln Y_t + (1 - \sigma) \ln t_{ij,t} - (1 - \sigma) \ln P_{j,t} - (1 - \sigma) \ln \Pi_{i,t} + \varepsilon_{ij,t}. \quad (1-11)$$

Hundreds of papers have estimated versions of (1-11) to study distance, RTAs, tariffs, currency unions, WTO membership, and much else. Yet, as this Part shows, gravity is still routinely applied a-theoretically, in ways that bias — sometimes badly — the resulting estimates. This Part works through eight estimation challenges, the literature's solutions to each, and distils them into six practical recommendations.

4 Lesson 4: Challenges 1–3 — multilateral resistances, zeros, and heteroscedasticity

Definition: Challenge 1 — Multilateral resistances are unobserved

$P_{j,t}$ and $\Pi_{i,t}$ in (1-11) are theoretical constructs, not directly observed by the researcher. Baldwin and Taglioni (2006) call ignoring them the “Gold Medal Mistake” in gravity estimation.

Derivation: Four solutions to Challenge 1, in order of sophistication

1. **Custom nonlinear iteration** (Anderson and van Wincoop, 2003): iterate between estimating trade costs and reconstructing $\{\Pi_i, P_j\}$ until convergence.
2. **Remoteness indexes**: proxy Π_i, P_j with GDP-weighted average distances (Wei, 1996; Baier and Bergstrand, 2009) — convenient, but Head and Mayer (2014) show these bear little resemblance to the theoretical multilateral resistances.
3. **Ratio methods** that algebraically cancel the multilateral resistances (Head and Ries, 2001; Novy, 2013) — covered in the general-equilibrium companion (Chapter 2).
4. **Directional fixed effects** (Hummels, 2001; Feenstra, 2016; Olivero and Yotov, 2012): include exporter(-time) and importer(-time) dummies. This is now the standard, and it simultaneously absorbs $Y_{i,t}$, $E_{j,t}$, and *any* other exporter- or importer-specific observable or unobservable.

Definition: Challenge 2 — Zero trade flows

OLS on $\ln X_{ij,t}$ silently drops every zero-trade observation, and the problem worsens with disaggregation (especially services).

Derivation: Solutions to Challenge 2

- *Ad hoc small constants*: theoretically inconsistent, results depend on units (Head and Mayer, 2014) — avoid.
- *Tobit* (Eaton and Tamura, 1995; Martin and Pham, 2008): gravity theory does not motivate a censoring threshold.
- *Two-step selection* (Helpman, Melitz and Rubinstein, 2008): a Probit for the extensive margin (fixed costs of entry) plus OLS on positive trade; hard to find valid exclusion restrictions.
- *Two-part model* (Egger et al., 2011): separates extensive and intensive margins, also addressing endogenous regressors like RTAs.
- *Poisson Pseudo Maximum Likelihood, PPML* (Santos Silva and Tenreyro, 2006): estimate gravity in **multiplicative** form. PPML naturally uses the information in zeros and performs well by Monte Carlo evidence even with many zeros. **This is the recommended default.**

Definition: Challenge 3 — Heteroscedasticity

Trade data are notoriously heteroscedastic. By Jensen's inequality, estimating the *log-linearized* gravity equation by OLS (or any estimator requiring a log transform) in the presence of heteroscedasticity yields estimates that are not just biased but **inconsistent** (Santos Silva and Tenreyro, 2006).

Derivation: Solutions to Challenge 3

(i) Estimate size-adjusted trade $X_{ij,t}/(E_{j,t}Y_{i,t})$ (Anderson and van Wincoop, 2003) — but this only addresses size-driven heteroscedasticity and does not solve Challenge 2. (ii) Apply **PPML** directly to the multiplicative gravity equation — this is both consistent under heteroscedasticity and solves the zeros problem simultaneously, which is why PPML dominates the toolkit going forward.

Intuition: Why PPML solves two problems with one estimator

PPML fits $X_{ij,t} = \exp(\cdot) \times \varepsilon_{ij,t}$ directly, without logging the dependent variable. Because zero is a perfectly valid value of $X_{ij,t}$ in levels (unlike $\ln 0$), the zeros stay in the estimation sample and contribute information. And because PPML's first-order conditions weight observations in a way that matches the variance structure implied by heteroscedastic multiplicative errors, the resulting estimates remain consistent even though OLS on the logged equation would not be. One estimator, two structural problems solved.

Understanding Quiz 4 — Lesson 4

Q4.1 Including exporter-time and importer-time fixed effects in a gravity regression accounts for:

- (A) bilateral tariffs only
- (B) the unobserved multilateral resistance terms (and country-specific size/characteristics)
- (C) random error only
- (D) nothing; they must be estimated separately

Q4.2 The main reason OLS on $\ln X_{ij}$ handles zero trade flows poorly is that:

- (A) zero-trade observations are mechanically dropped when trade is logged
- (B) zero trade never occurs in real data
- (C) OLS cannot include fixed effects
- (D) PPML cannot be estimated on positive trade

Q4.3 Under heteroscedasticity, log-linear OLS estimates of the gravity model are:

- (A) unbiased and consistent
- (B) biased but consistent
- (C) biased and inconsistent, by Jensen's inequality
- (D) always identical to PPML

Q4.4 PPML is attractive relative to OLS mainly because it:

- (A) requires fewer observations
- (B) simultaneously handles zero trade flows and heteroscedasticity-driven inconsistency
- (C) ignores multilateral resistances by design
- (D) only works with cross-sectional data

5 Lesson 5: Challenges 4–8 — trade costs, endogeneity, and non-discriminatory policy

Definition: Challenge 4 — Specifying bilateral trade costs

The unobserved term $(1 - \sigma) \ln t_{ij,t}$ in (1-11) is usually proxied by a standard battery of covariates:

$$(1 - \sigma) \ln t_{ij,t} = \beta_1 \ln DIST_{ij} + \beta_2 CNTG_{ij} + \beta_3 LANG_{ij} + \beta_4 CLNY_{ij} + \beta_5 RTA_{ij,t} + \beta_6 \tau_{ij,t} \quad (1-12)$$

with $\tau_{ij,t} = \ln(1 + tariff_{ij,t})$. Because tariffs are direct price shifters, their coefficient recovers the trade elasticity directly: $\hat{\sigma} = -\hat{\beta}_6$.

Definition: Challenge 5 — Endogeneity of trade policy

$RTA_{ij,t}$ and $\tau_{ij,t}$ are plausibly endogenous: countries are more likely to sign an RTA with a country they *already* trade heavily with (reverse causality).

Derivation: The pair fixed-effects solution (Baier and Bergstrand, 2007)

With panel data, adding **country-pair fixed effects** μ_{ij} eliminates (or absorbs) the unobserved, time-invariant linkages between trade policy and the gravity error. Pair fixed effects also absorb *all* time-invariant standard gravity covariates (distance, contiguity, language, colony) — which is not a loss, because trade policy is time-varying by construction and remains identified. Egger and Nigai (2015) and Agnosteva et al. (2014) show pair fixed effects are in fact a *better* measure of bilateral trade costs than the standard covariate list.

Definition: Challenge 6 — Non-discriminatory trade policy

Unilateral policies (export subsidies, MFN tariffs) are exporter- or importer-specific, so they are *mechanically absorbed* by the exporter-time/importer-time fixed effects required to handle Challenge 1 — they cannot be estimated in the standard specification.

Derivation: The intra-national trade solution (Heid et al., 2015)

Interact the country-specific policy with a dummy $INTL_{ij}$ (one for international, zero for intra-national trade): $NES_{i,t} \times INTL_{ij}$ and $NIP_{j,t} \times INTL_{ij}$ become genuinely *bilateral* covariates, escaping collinearity with the directional fixed effects. This trick only works if intra-national trade flows are included in the estimation sample — domestic trade is the reference point against which international policy effects are measured.

Definition: Challenge 7 — Adjustment lags; Challenge 8 — Disaggregated/sectoral gravity

Challenge 7: trade does not adjust to policy instantaneously, and pooling consecutive years overstates precision (Trefler, 2004). **Solution:** use *interval* panels (Olivero and Yotov, 2012 show 3-, 4-, and 5-year intervals give similar estimates). **Challenge 8:** many policies (tariffs)

are sectoral. **Solution:** gravity is *separable* (Anderson and van Wincoop, 2004; Larch and Yotov, 2016b), so the sectoral gravity equation

$$X_{ij,t}^k = \frac{Y_{i,t}^k E_{j,t}^k}{Y_t^k} \left(\frac{t_{ij,t}^k}{P_{j,t}^k \Pi_{i,t}^k} \right)^{1-\sigma^k} \quad (1-13)$$

can be estimated sector by sector exactly as if the data were aggregate.

Practical Recommendation: The six recommendations, in one place

1. Use **panel** data whenever available.
2. Use panel data with **intervals** (not consecutive years).
3. Include **intra-national** trade flows alongside international trade.
4. Include **exporter-time and importer-time fixed effects**.
5. Include **pair fixed effects** in panel estimations.
6. Estimate with **PPML**.

Understanding Quiz 5 — Lesson 5

Q5.1 Since tariffs are direct price shifters, the coefficient on $\tau_{ij,t}$ in (1-12) recovers:

- (A) the distance elasticity
- (B) minus the trade elasticity of substitution, $\hat{\sigma} = -\hat{\beta}_6$
- (C) the RTA effect
- (D) nothing useful

Q5.2 Country-pair fixed effects address endogeneity of RTAs because they:

- (A) are correlated with GDP only
- (B) absorb unobserved, time-invariant bilateral factors correlated with both trade policy and trade costs
- (C) eliminate the need for any trade cost proxy
- (D) only apply to cross-sectional data

Q5.3 Non-discriminatory policies such as MFN tariffs are hard to estimate in standard gravity specifications because:

- (A) they never vary over time
- (B) they are absorbed by the exporter-time/importer-time fixed effects needed to control for multilateral resistances
- (C) they are always zero
- (D) PPML cannot estimate them

Q5.4 The separability property of structural gravity implies that sectoral gravity equations can be:

- (A) never estimated separately from the aggregate
- (B) estimated sector by sector exactly as if the data were aggregate
- (C) estimated only with cross-sectional data
- (D) estimated only for goods, never services

6 Lesson 6: The comprehensive, theory-consistent estimating equation

Derivation: Putting it all together

Combining all six recommendations delivers the generic, comprehensive specification:

$$X_{ij,t} = \exp \left[\pi_{i,t} + \chi_{j,t} + \mu_{ij} + \eta_1 BTP_{ij,t} + \eta_2 NES_{i,t} \times INTL_{ij} + \eta_3 NIP_{j,t} \times INTL_{ij} \right] \times \varepsilon_{ij,t} \quad (1-14)$$

where $\pi_{i,t}$ (exporter-time FE) absorbs the outward multilateral resistance and any exporter-specific shock; $\chi_{j,t}$ (importer-time FE) absorbs the inward multilateral resistance and any importer-specific shock; μ_{ij} (pair FE) absorbs all time-invariant bilateral trade costs and addresses endogeneity; $BTP_{ij,t}$ is any *bilateral* trade policy (RTAs, bilateral tariffs, currency unions); and $NES_{i,t} \times INTL_{ij}$, $NIP_{j,t} \times INTL_{ij}$ identify non-discriminatory export-support and import-protection policies, respectively, thanks to the intra-national trade trick of Lesson 5.

Intuition: Reading μ_{ij} 's dual role

Pair fixed effects do two jobs at once. First, statistically, they are the most flexible possible measure of time-invariant bilateral trade costs — more flexible than any finite list of covariates like distance or common language, because they absorb *everything* time-invariant about the pair, observed or not. Second, econometrically, they are what makes $\varepsilon_{ij,t}$ credible as a genuine measurement error rather than a repository of omitted, policy-correlated trade costs. Whether $\varepsilon_{ij,t}$ enters additively or multiplicatively does not matter for PPML (Santos Silva and Tenreyro, 2006) — another reason PPML is the estimator of choice.

Understanding Quiz 6 — Lesson 6

Q6.1 In equation (1-14), the term that absorbs the outward multilateral resistance is:

- (A) μ_{ij}
- (B) $\pi_{i,t}$
- (C) $\chi_{j,t}$
- (D) $\varepsilon_{ij,t}$

Q6.2 The interaction $NIP_{j,t} \times INTL_{ij}$ identifies the effect of a non-discriminatory import policy because:

- (A) it varies only across exporters
- (B) it is bilateral by construction and therefore escapes collinearity with importer-time fixed effects, once intra-national trade is in the sample
- (C) it is always statistically insignificant
- (D) it replaces the need for pair fixed effects

Q6.3 Whether the error term in the multiplicative gravity model enters additively or multiplicatively:

- (A) drastically changes the OLS estimates
- (B) does not matter for the PPML estimator
- (C) makes zero trade flows impossible to estimate
- (D) is irrelevant to the specification of fixed effects

Part III

Interpreting, Aggregating, and Sourcing Gravity Estimates

7 Lesson 7: Interpreting gravity estimates — trade volumes and tariff equivalents

7.1 Trade volume effects

Derivation: Continuous vs. indicator covariates

For a **continuous** covariate in logs (e.g. $\ln DIST_{ij}$), the coefficient is directly an elasticity: $\hat{\beta}_{DIST} = -1$ means a 10% increase in distance lowers trade by about 10%. For an **indicator** covariate (e.g. $RTA_{ij,t}$), the percentage trade-volume effect is

$$(e^{\hat{\beta}_{dummy}} - 1) \times 100. \quad (1-16)$$

For example, Baier and Bergstrand's (2007) benchmark $\hat{\beta}_{RTA} = 0.76$ implies RTAs formed between 1960 and 2000 raised trade on average by $(e^{0.76} - 1) \times 100 \approx 114\%$.

Intuition: Structural vs. trade-cost components

Except for direct price-shifters like tariffs, most gravity coefficients bundle (i) a *structural* component and (ii) a *trade-cost elasticity*. For instance $\hat{\beta}_{DIST} = (1 - \hat{\sigma})\rho$, where ρ is the elasticity of trade costs with respect to distance. This decomposition matters twice: it lets researchers back out $\rho = \hat{\beta}_{DIST}/(1 - \hat{\sigma})$ — showing that “distance” in gravity regressions captures far more than freight costs (Head and Mayer, 2013) — and it is the key to converting *any* estimate into a tariff equivalent, next.

7.2 Tariff-equivalent effects

Derivation: Converting any gravity estimate into a tariff equivalent

The tariff-equivalent effect of an RTA is the ad-valorem tariff cut that would have produced the same trade effect:

$$(e^{\hat{\beta}_{RTA} - \hat{\beta}_{TARIFF}} - 1) \times 100. \quad (1-17)$$

Since $\hat{\beta}_{TARIFF} = -\hat{\sigma}$ structurally, this becomes

$$(e^{\hat{\beta}_{RTA}/\hat{\sigma}} - 1) \times 100, \quad (1-18)$$

which requires *no tariff data at all* — only an external estimate of σ . With $\hat{\beta}_{RTA} = 0.76$ and a representative $\sigma = 5$, the RTA tariff-equivalent is $(e^{0.76/5} - 1) \times 100 \approx 16.4\%$.

Understanding Quiz 7 — Lesson 7

Q7.1 For a continuous log covariate such as distance, the gravity coefficient is interpreted directly as:

- (A) a probability
- (B) an elasticity of trade with respect to that variable
- (C) a tariff rate
- (D) a fixed cost

Q7.2 The trade volume effect of an indicator variable with estimate $\hat{\beta}_{dummy}$ is:

- (A) $\hat{\beta}_{dummy} \times 100$
- (B) $(e^{\hat{\beta}_{dummy}} - 1) \times 100$
- (C) $1/\hat{\beta}_{dummy}$
- (D) $e^{-\hat{\beta}_{dummy}}$

Q7.3 Computing the tariff equivalent of an RTA via equation (1-18) requires:

- (A) observed bilateral tariff data for the RTA
- (B) only the RTA coefficient and an estimate of the trade elasticity σ
- (C) a computable general equilibrium model
- (D) nothing; it is always zero

8 Lesson 8: Consistent aggregation of bilateral trade costs

Intuition: Why aggregate trade costs at all?

A customs union may want to know its overall trade-cost stance toward the rest of the world; a country may want to aggregate interprovincial costs to the national level; a statistical agency may want one number for a whole sector. Simple averages of t_{ij} are *not* theoretically consistent — they ignore how trade actually responds to costs. The structural gravity model supplies the right weights.

Derivation: The volume-equivalent uniform trade cost, $b_{CU}(i)$

Fix a subset of destinations forming a customs union $CU(i)$. The uniform bilateral cost $b_{CU}(i)$ that *preserves* i 's aggregate export volume to $CU(i)$ solves

$$\sum_{j \in CU(i)} X_{ij} = \sum_{j \in CU(i)} \frac{Y_i E_j}{Y} \left(\frac{t_{ij}}{\Pi_i P_j} \right)^{1-\sigma} = \sum_{j \in CU(i)} \frac{Y_i E_j}{Y} \left(\frac{b_{CU}(i)}{\Pi_i P_j} \right)^{1-\sigma}. \quad (1-19)$$

Solving for $b_{CU}(i)$:

$$b_{CU}(i) = \left[\sum_{j \in CU(i)} \frac{E_j / P_j^{1-\sigma}}{\sum_{k \in CU(i)} E_k / P_k^{1-\sigma}} t_{ij}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (1-20)$$

Intuition: A market-access weighted average

Equation (1-20) says: the consistent aggregate trade cost is a **weighted average of the bilateral costs**, where the weight on destination j is j 's share of $CU(i)$ -wide “market potential” $E_j/P_j^{1-\sigma}$ — exactly the kind of index familiar from the economic-geography literature (Redding and Venables, 2004). Practically, once the gravity model (1-14) is estimated by PPML, the weights come *for free* from the importer fixed effects:

$$b_{CU}(i) = \left[\sum_{j \in CU(i)} \frac{\chi_j}{\sum_{k \in CU(i)} \chi_k} t_{ij}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (1-21)$$

A symmetric, demand-side aggregate $b_{CU}(j)$ uses the *exporter* fixed effects π_i as weights (1-22). This works precisely because PPML's fixed effects correspond *exactly* to the structural terms Π_i, P_j (Arvis and Shepherd, 2013; Fally, 2015) — a recurring theme in this Guide.

Derivation: Three practical steps

1. Estimate the structural gravity model with PPML.
2. Construct bilateral trade costs $\hat{t}_{ij,t}$ for each pair.
3. Aggregate using importer fixed effects as weights for supply-side aggregation, or exporter fixed effects for demand-side aggregation.

Understanding Quiz 8 — Lesson 8

Q8.1 The consistent aggregate trade cost $b_{CU}(i)$ is defined so as to preserve:

- (A) i 's aggregate export volume to the customs union
- (B) the simple average of bilateral costs
- (C) the elasticity of substitution
- (D) nothing observable

Q8.2 The weights in the consistent aggregation formula (1-20) are:

- (A) equal weights across all destinations
- (B) each destination's share of market potential, $E_j/P_j^{1-\sigma}$, within the union
- (C) random
- (D) based on physical distance only

Q8.3 In practice, the aggregation weights in equation (1-21) can be recovered directly from:

- (A) survey data
- (B) the PPML importer fixed effects estimated in the gravity regression
- (C) GDP deflators
- (D) tariff schedules

9 Lesson 9: Gravity data — sources, coverage, and limitations

Definition: Trade flow data

Aggregate: IMF Direction of Trade Statistics (DOTS, 184 countries, annual from 1947). **Merchandise/disaggregated:** UN COMTRADE (from 1962, ~160+ countries, HS 6-digit or SITC classifications); CEPII's BACI cleans and reconciles COMTRADE mirror data; Feenstra and Romalis's World Trade Flows (WTF) database screens out implausible c.i.f./f.o.b. ratios. **Services:** OECD Trade in Services (35 countries from 1999); UN Service Trade Database; WTO-UNCTAD-ITC joint database. **Agriculture:** FAOSTAT Detailed Trade Matrix. **Intra-national trade:** constructed as gross production minus total exports; requires matching value-added production data (SIC/ISIC classifications) to gross export data — available via the World Bank's TPP database, CEPII's TradeProd, and UNIDO's INDSTAT.

Intuition: Why import data, and why “mirroring,” are usually preferred

Import data tend to be more reliable than export data because customs authorities monitor imports more closely (many are subject to duties). “Mirroring” — using destination j 's reported imports from i as the measure of i 's exports to j — is standard practice, though it can misfire when the importer has high tariffs and weak customs enforcement, since under-declaration then biases mirrored imports downward relative to true exports.

Definition: Trade cost data

Time-invariant/geographic: CEPII's GeoDist (225 countries) — physical distance, contiguity, common language, colonial history. **Trade policy:** tariffs and non-tariff measures, divided broadly into applied bilateral preferential tariffs (RTA schedules) and MFN tariffs (WTO/UNCTAD TRAINS, WITS).

Understanding Quiz 9 — Lesson 9

Q9.1 Import data are generally preferred to export data in gravity estimation because:

- (A) exports are never reported
- (B) imports are typically monitored more closely by customs authorities, often due to duties
- (C) imports are always larger in value
- (D) export data do not exist before 1995

Q9.2 Constructing intra-national trade flows typically requires:

- (A) bilateral tariff schedules only
- (B) the difference between gross production (value added) and total exports, requiring matched production and trade classifications
- (C) CES preference parameters
- (D) nothing beyond COMTRADE data

Part IV

Applications: Traditional Estimates, the Distance Puzzle, and RTAs

10 Lesson 10: Traditional gravity estimates — OLS, remoteness, fixed effects, PPML

Derivation: Four horse-race specifications

Using 4-year interval panel data (1986–2006), four increasingly rigorous specifications are estimated:

- (1) OLS, no MR control: $\ln X_{ij,t} = \beta_0 + \beta_1 \ln DIST_{ij} + \beta_2 CNTG_{ij} + \beta_3 LANG_{ij} + \beta_4 CLNY_{ij} + \beta_5 \ln Y_{ij,t}$
- (2) OLS, remoteness indexes: adds $\ln REM_EXP_{i,t}$, $\ln REM_IMP_{j,t}$
- (3) OLS, directional FE: $\ln X_{ij,t} = \pi_{i,t} + \chi_{j,t} + \beta_1 \ln DIST_{ij} + \beta_2 CNTG_{ij} + \beta_3 LANG_{ij} + \beta_4 CLNY_{ij} + \beta_5 \ln Y_{ij,t}$
- (4) PPML, directional FE: $X_{ij,t} = \exp[\pi_{i,t} + \chi_{j,t} + \beta_1 \ln DIST_{ij} + \beta_2 CNTG_{ij} + \beta_3 LANG_{ij} + \beta_4 CLNY_{ij} + \beta_5 \ln Y_{ij,t}]$

	(1) OLS	(2) OLS Remoteness	(3) OLS FE	(4) PPML FE
Log distance	−1.002**	−1.185**	−1.216**	−0.841**
Contiguity	0.574**	0.247	0.223	0.437**
Common language	0.802**	0.739**	0.661**	0.247**
Colony	0.735**	0.842**	0.670**	−0.222 ⁺
R^2	0.759	0.765	0.843	0.614
RESET test (p -value)	0.000	0.000	0.000	0.642
Exporter/importer-time FE	No	No	Yes	Yes

Table 1: Traditional gravity estimates (abbreviated from Table 1). Standard errors clustered by pair.

Intuition: What the horse race teaches

Three lessons emerge. **First**, distance is a robust, powerful trade deterrent throughout, close to the benchmark -1 found across the literature (Disdier and Head, 2008). **Second**, moving from (1) to (3) — from no multilateral-resistance control, to remoteness indexes, to full directional fixed effects — *changes* the coefficients substantially (distance strengthens, contiguity and language weaken), exactly the bias Anderson and van Wincoop (2003) warned about. **Third**, and most important, only the **PPML** specification (4) passes the Ramsey RESET misspecification test ($p = 0.642$); all three OLS variants are rejected. This is direct evidence, not just theoretical argument, in favour of PPML.

Understanding Quiz 10 — Lesson 10

Q10.1 Moving from specification (1) to specification (3) in Table 1 changes the distance and language coefficients because:

- (A) the sample changes
- (B) specification (1) fails to control for the multilateral resistance terms, biasing its estimates
- (C) PPML is used in (3)
- (D) distance data changed over time

Q10.2 Which specification in Table 1 passes the RESET misspecification test?

- (A) (1) OLS
- (B) (2) OLS with remoteness
- (C) (3) OLS with fixed effects
- (D) (4) PPML with fixed effects

11 Lesson 11: The “distance puzzle” resolved

Intuition: The puzzle

Disdier and Head’s (2008) meta-analysis of 1,467 distance estimates finds the negative impact of distance on trade has *not* declined over time — indeed it appears to have *risen* — flatly contradicting the observed decline in transport and communication costs (“the distance puzzle”). Yotov (2012) resolves it: because structural gravity can only ever identify *relative* trade costs, estimating distance effects using **international trade data only** measures international distance costs *relative to other international costs* — not relative to the true benchmark, internal (intra-national) trade costs.

Derivation: Uncovering, then solving, the puzzle

Step 1 (uncovering it). With year-specific distance coefficients and international trade only,

$$\ln X_{ij,t} = \pi_{i,t} + \chi_{j,t} + \sum_{T=1986}^{2006} \beta_T \ln DIST(T)_{ij} + \beta_2 CNTG_{ij} + \beta_3 LANG_{ij} + \beta_4 CLNY_{ij} + \varepsilon_{ij,t} \quad (1-29)$$

the estimated distance penalty **risks** by 7.95% (OLS) or is statistically flat at -2.75% (PPML) between 1986 and 2006 — the puzzle, confirmed.

Step 2 (solving it). Add intra-national trade and intra-national distance $\ln DIST_INTRA_{ii}$ (zero for international pairs):

$$X_{ij,t} = \exp \left[\pi_{i,t} + \chi_{j,t} + \sum_T \beta_T \ln DIST(T)_{ij} + \beta_2 CNTG_{ij} + \beta_3 LANG_{ij} + \beta_4 CLNY_{ij} + \beta_5 \ln DIST_INTRA_{ii} \right] \times \varepsilon_{ij,t} \quad (1-31)$$

Now international distance’s estimated effect **falls** by 10.97% over the same period — the puzzle dissolves once distance is measured relative to the correct internal benchmark.

	(1) OLS Intl. only	(2) PPML Intl. only	(3) +Intra dist.	(4) +Intra dummy	(5) +Intra FEs
Log distance 1986	−1.168	−0.859	−0.980	−0.857	−0.910
Log distance 2006	−1.261	−0.836	−0.872	−0.754	−0.811
% change 1986–2006	+7.95*	−2.75	−10.97**	−11.97**	−10.93**

Table 2: The distance puzzle, uncovered (columns 1–2) and solved (columns 3–5); abbreviated from Table 2.

Intuition: The role of the “home bias” dummy

Specification (4) adds $SMCTRY_{ii}$, a dummy equal to one for intra-national trade. Its estimated coefficient of 1.689 implies intra-national trade is $\exp(1.689) \approx 5.5$ times larger than a comparable international flow — a large, but far more modest, “border effect” than McCallum’s (1995) famous estimate of 22 for Canadian provinces vs. US states, once the multilateral resistances are properly controlled for (Anderson and van Wincoop, 2003).

Understanding Quiz 11 — Lesson 11

Q11.1 The “distance puzzle” refers to the empirical finding that:

- (A) distance has no effect on trade
- (B) the estimated negative effect of distance on *international* trade has not declined despite falling transport costs
- (C) RTAs have no effect on trade
- (D) trade costs are zero in the modern world

Q11.2 Yotov (2012) resolves the puzzle by measuring the effects of distance and globalization:

- (A) relative to a random reference country
- (B) relative to internal (intra-national) trade costs, by including domestic trade flows in the sample
- (C) using only cross-sectional data
- (D) by dropping distance from the regression

Q11.3 The estimated home-bias/intra-national trade dummy of 1.689 implies domestic trade is about:

- (A) 1.7 times larger than comparable international trade
- (B) 5.5 times larger than comparable international trade
- (C) 22 times larger than comparable international trade
- (D) equal to international trade

12 Lesson 12: Regional trade agreement effects — endogeneity, phasing, and globalization

Derivation: Six successive specifications

1. **Traditional OLS** (international trade only): $\hat{\beta}_{RTA} = -0.004$, statistically *zero* — RTAs appear to do nothing, a red flag for endogeneity bias.
2. **PPML** (international only): $\hat{\beta}_{RTA} = 0.191$ (+21% trade).

3. + **Intra-national trade** (trade diversion from domestic sales, Dai et al., 2014): $\hat{\beta}_{RTA} = 0.409$ (+51%) — more than double, confirming RTAs divert trade from *domestic* sales, not just from non-members.
4. + **Pair fixed effects** (endogeneity, Baier and Bergstrand, 2007): $\hat{\beta}_{RTA} = 0.557$ ($\approx +75\%$) — much larger, confirming that failing to address endogeneity **biases the RTA effect downward**.
5. + **RTA lead** (strict exogeneity test): $RTA_{ij,t+4}$ is insignificant (0.077), confirming no “reverse causality” once pair FEs are included.
6. + **Lagged RTAs** (phasing-in, up to 12 years) **and globalization controls** (international border dummies by year, Bergstrand et al., 2015): the total (summed) RTA effect is *cut roughly in half* once time-varying globalization forces are accounted for, and the estimated international-border penalty itself shrinks steadily over 1986–2006 — e.g. $\exp(0.706) \approx 2.03$ times larger in 1986 than in 2006 — direct evidence of globalization eroding border effects.

	(1) OLS	(2) PPML	(3) +Intra	(4) +Pair FE	(5) +Lead	(7) +Globlzn.
RTA	−0.004	0.191**	0.409**	0.557**	0.520**	0.116
Total RTA effect						0.475**

Table 3: Estimating the effects of RTAs, successively refined; abbreviated from Table 3.

Intuition: The big picture across the six specifications

Each refinement corrects a specific bias identified in Lessons 4–5: multilateral resistances (moving to PPML with directional FE), trade diversion from domestic sales (adding intra-national trade), and endogeneity (adding pair FE and testing strict exogeneity with a lead). The final, most defensible estimate — correcting for globalization trends that would otherwise be wrongly attributed to RTAs — is about half the naive PPML estimate, illustrating how much a theory-consistent specification can move a policy-relevant number.

Understanding Quiz 12 — Lesson 12

Q12.1 The RTA coefficient more than doubles when intra-national trade is added to the sample (specification 3) because:

- (A) of a coding error
- (B) RTAs divert trade away from *domestic* sales in addition to increasing trade with new partners, a source of bias when domestic trade is excluded
- (C) RTAs only affect intra-national trade
- (D) PPML requires intra-national trade to run at all

Q12.2 Adding pair fixed effects roughly doubles the RTA coefficient again (specification 4) because:

- (A) pair FE remove all variation
- (B) without pair FE, RTA estimates are biased downward due to endogeneity/reverse causality
- (C) pair FE increase the sample size

(D) RTAs become exogenous only under OLS

Q12.3 The RTA lead test ($RTA_{ij,t+4}$) is used to check:

(A) whether RTAs phase in over 12 years

(B) whether pair fixed effects have successfully addressed reverse causality (strict exogeneity)

(C) the trade elasticity of substitution

(D) the distance puzzle

Q12.4 Once globalization (time-varying international border effects) is controlled for, the total estimated RTA effect:

(A) stays exactly the same

(B) roughly doubles

(C) is cut roughly in half, since part of the earlier RTA estimate was capturing generic globalization trends

(D) becomes negative

Final Exam: Comprehensive Multiple-Choice Quiz

Understanding Quiz — Final Exam (all parts)

F1. In the Newton/gravity analogy, the gravitational constant G maps to:

(A) the elasticity of substitution

(B) the inverse of world output, $1/Y$

(C) bilateral distance

(D) the trade elasticity θ

F2. The structural gravity equation $X_{ij} = (Y_i E_j / Y)(t_{ij} / (\Pi_i P_j))^{1-\sigma}$ is derived by combining CES demand with:

(A) free entry

(B) market clearing for each origin's output

(C) balanced trade for each pair

(D) fixed wages

F3. The outward and inward multilateral resistances Π_i, P_j are:

(A) exogenous policy parameters

(B) jointly, simultaneously determined general-equilibrium trade cost aggregates

(C) equal to bilateral distance

(D) irrelevant for gravity estimation

F4. The Eaton–Kortum supply-side derivation of gravity replaces $1 - \sigma$ with:

(A) $-\theta$, the Fréchet dispersion parameter

(B) the number of goods

(C) the labour share

(D) zero

F5. PPML is preferred to OLS on logged gravity mainly because it:

(A) is always faster to compute

- (B) remains consistent under heteroscedasticity and uses information in zero trade flows
- (C) requires fewer covariates
- (D) cannot include fixed effects

F6. Country-pair fixed effects in a panel gravity regression:

- (A) identify the effect of bilateral distance
- (B) absorb time-invariant bilateral trade costs and help address the endogeneity of trade policy
- (C) are redundant with exporter-time fixed effects
- (D) cannot be used with PPML

F7. Non-discriminatory policies (e.g. MFN tariffs) can be identified in a gravity model with directional fixed effects by:

- (A) dropping the fixed effects
- (B) interacting the policy with an international-trade dummy and including intra-national trade in the sample
- (C) using cross-sectional data only
- (D) averaging tariffs across all partners

F8. The tariff-equivalent of an RTA can be computed from $\hat{\beta}_{RTA}$ alone, without tariff data, using:

- (A) $\hat{\beta}_{RTA} \times 100$
- (B) $(e^{\hat{\beta}_{RTA}/\sigma} - 1) \times 100$, given an external estimate of σ
- (C) the distance coefficient
- (D) nothing; tariff data are always required

F9. Consistent aggregation of bilateral trade costs to the customs-union level weights each bilateral cost by:

- (A) equal weights
- (B) each destination's share of market potential within the union
- (C) population only
- (D) nothing; simple averages are theoretically consistent

F10. Import data are typically preferred over export data in constructing the gravity dependent variable because:

- (A) exports are never recorded
- (B) imports are more closely monitored by customs, often due to duties
- (C) imports are always larger
- (D) export data are classified differently every year

F11. The “distance puzzle” (the apparent lack of decline in the distance coefficient over time) is resolved once:

- (A) distance is dropped from the regression
- (B) intra-national trade and intra-national distance are included, so that international distance is measured relative to the correct internal benchmark
- (C) PPML is replaced with OLS
- (D) colonial ties are excluded

F12. Compared to specification without pair fixed effects, adding pair fixed effects to the RTA gravity regression:

- (A) lowers the estimated RTA effect due to overfitting
- (B) raises the estimated RTA effect, consistent with downward endogeneity bias in its absence

- (C) has no effect on the RTA coefficient
- (D) eliminates the RTA variable entirely

F13. Once time-varying globalization forces (international border dummies) are controlled for, the estimated total RTA effect in the applications of this chapter:

- (A) increases substantially
- (B) is roughly cut in half
- (C) becomes negative and significant
- (D) is unchanged

F14. The separability property of the structural gravity model implies that:

- (A) sectoral gravity equations cannot be estimated
- (B) sectoral (or regional) gravity equations take exactly the same form as the aggregate equation and can be estimated the same way
- (C) only aggregate data can ever be used
- (D) trade costs must be identical across sectors

F15. The Arkolakis–Costinot–Rodríguez-Clare isomorphism result implies that:

- (A) only the Armington model is theoretically valid
- (B) many structurally different trade models (Armington, Ricardian/EK, heterogeneous firms, Heckscher-Ohlin) generate the same gravity equation and the same gains-from-trade formula
- (C) gravity fails empirically
- (D) comparative advantage is irrelevant to gravity

A Answer key

Lesson quizzes

Lesson 1

Q1.1: C. $T_{ij}^\theta \equiv (t_{ij}(\Pi_i P_j))^{\sigma-1}$ bundles the direct bilateral cost and both multilateral resistances. **Q1.2:** B. Tinbergen (1962) and earlier applications were atheoretical curve-fitting. **Q1.3:** B. ACR (2012) show isomorphism across a wide class of structurally different trade models.

Lesson 2

Q2.1: B. Armington (1969): differentiation by place of origin. **Q2.2:** B. $X_{ij} \propto E_j \cdot P_j^{\sigma-1}$: increasing in both. **Q2.3:** B. $\Pi_i^{1-\sigma} = \sum_j (t_{ij}/P_j)^{1-\sigma} (E_j/Y)$: a destination-expenditure-weighted aggregate. **Q2.4:** B. Π_i and P_j solve simultaneously for all i, j at once.

Lesson 3

Q3.1: B. Setting $t_{ij} = 1$ eliminates the trade-cost term, isolating $Y_i E_j / Y$. **Q3.2:** B. They summarize each country's market access relative to the rest of the world. **Q3.3:** A. $-\theta$ is the Fréchet dispersion (comparative-advantage scope) parameter. **Q3.4:** B. Aggregate trade data cannot distinguish which of the isomorphic models generated it.

Lesson 4

Q4.1: B. Directional (exporter-time/importer-time) FE absorb $\Pi_{i,t}$, $P_{j,t}$, $Y_{i,t}$, $E_{j,t}$ and more. **Q4.2:** A. $\ln 0$ is undefined, so zero-trade rows drop out of an OLS-on-logs regression. **Q4.3:** C. Jensen's inequality implies inconsistency, not just bias, under heteroscedasticity with a log transform. **Q4.4:** B. PPML solves the zeros problem and the heteroscedasticity-inconsistency problem simultaneously.

Lesson 5

Q5.1: B. Tariffs are direct price-shifters, so $\hat{\sigma} = -\hat{\beta}_6$. **Q5.2:** B. Pair FE absorb time-invariant unobserved links between policy and trade costs. **Q5.3:** B. Collinearity with the directional FE required for Challenge 1. **Q5.4:** B. Separability: sectoral gravity has the exact same functional form as aggregate gravity.

Lesson 6

Q6.1: B. $\pi_{i,t}$ is the exporter-time fixed effect. **Q6.2:** B. The interaction with $INTL_{ij}$, combined with intra-national trade in the sample, creates bilateral variation. **Q6.3:** B. PPML's consistency does not depend on the additive/multiplicative form of $\varepsilon_{ij,t}$.

Lesson 7

Q7.1: B. A log-log coefficient is an elasticity. **Q7.2:** B. $(e^{\hat{\beta}_{dummy}} - 1) \times 100$. **Q7.3:** B. Equation (1-18) needs only $\hat{\beta}_{RTA}$ and an external σ .

Lesson 8

Q8.1: A. $b_{CU}(i)$ is defined to preserve i 's aggregate export volume to the union. **Q8.2:** B. Weights are each destination's share of union-wide market potential $E_j/P_j^{1-\sigma}$. **Q8.3:** B. PPML importer fixed effects deliver the weights directly.

Lesson 9

Q9.1: B. Customs monitoring (duties) makes import data more reliable. **Q9.2:** B. Intra-national trade = gross production (value added) minus total exports.

Lesson 10

Q10.1: B. Specification (1) fails to control for multilateral resistances, biasing estimates. **Q10.2:** D. Only PPML with fixed effects (column 4) passes RESET, $p = 0.642$.

Lesson 11

Q11.1: B. The negative distance effect on international trade fails to decline despite falling real transport costs. **Q11.2:** B. Yotov (2012) benchmarks against intra-national trade costs. **Q11.3:** B. $\exp(1.689) \approx 5.5$.

Lesson 12

Q12.1: B. RTAs also divert trade from domestic sales; excluding intra-national trade masks this. **Q12.2: B.** Without pair FE, endogeneity biases the RTA coefficient downward (Baier and Bergstrand, 2007). **Q12.3: B.** A significant lead would indicate reverse causality remains; the insignificant lead confirms strict exogeneity. **Q12.4: C.** Part of the naive RTA estimate was capturing generic globalization/border-cost decline.

Final exam

Final Exam key

- F1: B.** $G \equiv 1/Y$.
- F2: B.** Market clearing, $Y_i = \sum_j X_{ij}$, closes the CES demand system into gravity.
- F3: B.** Π_i, P_j solve jointly as a general-equilibrium system.
- F4: A.** EK's Fréchet shape parameter θ replaces $\sigma - 1$.
- F5: B.** PPML handles heteroscedasticity-driven inconsistency and zero trade jointly.
- F6: B.** Pair FE absorb time-invariant costs and address endogeneity.
- F7: B.** Interacting with $INTL_{ij}$ plus intra-national trade data.
- F8: B.** $(e^{\hat{\beta}_{RTA}/\sigma} - 1) \times 100$ needs only σ .
- F9: B.** Market-potential weights, $E_j/P_j^{1-\sigma}$.
- F10: B.** Import data face closer customs scrutiny.
- F11: B.** Including intra-national trade/distance resolves the puzzle.
- F12: B.** Endogeneity biases the naive RTA estimate downward; pair FE correct this.
- F13: B.** The total effect roughly halves once globalization trends are controlled for.
- F14: B.** Separability: identical functional form at the sectoral level.
- F15: B.** A wide model class shares the same gravity reduced form and gains-from-trade formula.